Subword Tokenization

- Subword tokenization is a middle ground between word-level and character-level tokenization

- First developed for machine translation -based on byte pair encoding (Gage, 1994)

Neural Machine Translation of Rare Words with Subword Units

Rico Sennrich and **Barry Haddow** and **Alexandra Birch**
School of Informatics, University of Edinburgh
{rico.sennrich,a.birch}@ed.ac.uk, bhaddow@inf.ed.ac.uk

The main motivation behind this paper is that the translation of some words is transparent in that they are translatable by a competent translator even if they are novel to him or her, based on a translation of known subword units such as morphemes or phonemes.

Subword Tokenization

- To avoid unknown words, we must add all possible characters/ symbols

- But, there are ~138K unicode symbols

- Instead, use bytes!
  - LLMs often use BPE with some rules to prevent certain types of merges
  - Modest-sized LLMs often have vocabulary sizes of 32-64K
  - With the big proprietary models, it is often ~100k–256k tokens

Subword Tokenization

- Other subword tokenization methods:
  - WordPiece (Schuster et al., 2012): merge to increase likelihood as measured by a language model (vs. frequency as in BPE)
  - SentencePiece (Kudo et al., 2018): can do subword tokenization without pre-tokenization (i.e., using white spaces) - good for languages without such word boundaries – Although pre-tokenization still often helps.